

Modeling Uncertain Demand with Random Variables

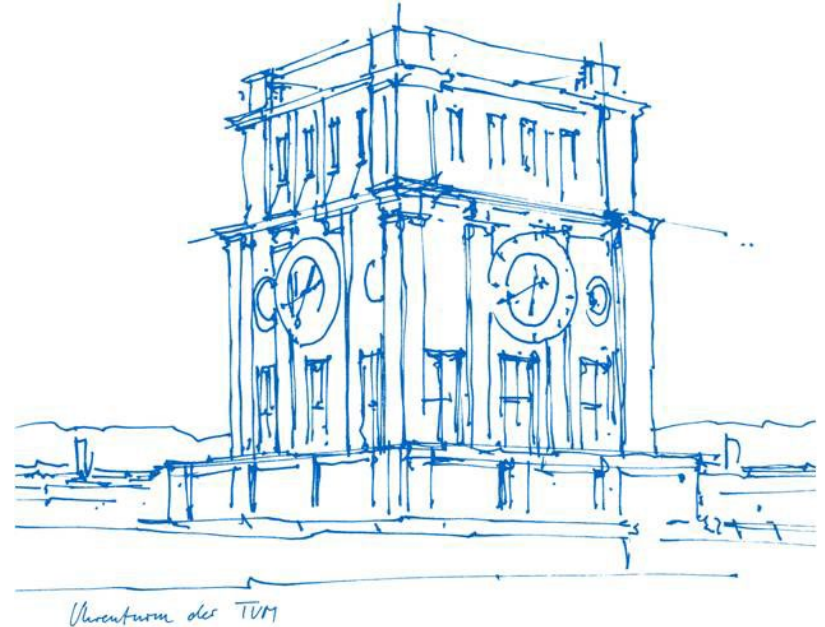
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Motivation

Objective

After this session, you will understand the concept of random variables and how they can be used to model uncertain demand.

You will learn how to calculate with random variables and their distributions so that you can apply them in the context of inventory management.

Random Variable

A random variable is a function (!) that assigns a value to an outcome of a random event.

Example 1: We assign the value “6” to the event that a dice is rolled and ends up with the side on top with six dots.

Random Variable

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Example 2: Mr Smith enters a store to buy 1 apple at 9:15 AM. Mr. Anderson buys 1 apple at 10:45 AM. Finally, Mrs. Trinity buys 1 apple at 2:30 PM. The total demand for apples in this time span is 3. Demand is a random variable which maps the event “customers enter a store at some time and buy apples” to a number. With this random number, we are neither interested *who* buys an apple nor *when* exactly on the day it happens. It is a simplified model of reality.

Random Variable

Random variables are usually capitalized, e.g. D for demand.

We can write them as if they were variables. E.g., if p represents price and D is a random variable representing demand then pD is the revenue.

Sample Space

Let X be a random variable.

The set of all possible realizations of X is called sample space and abbreviated with Ω .

We call subsets of Ω events, typically referred to with variable A .

Example:

The random variable “number of dots on a dice” has sample space $\Omega = \{1, 2, 3, 4, 5, 6\}$. The event “odd number of dots shown” is $A = \{1, 3, 5\}$ and clearly $A \subset \Omega$.

Probability

Random Variables and Probability

We are usually interested in the probability of outcomes. E.g., how likely will the demand be less or equal to 10 apples per day?

We often assign probabilities to outcomes; informally, we can write $Prob(D \in \{2\})$ to indicate the probability that demand is 2. Sometimes we simply write instead $Prob(D = 2)$.

We can write $Prob(D \in [2,4])$ to indicate the probability that demand is between 2 and 4. Or, with any set A of possible demand realizations we can write $Prob(D \in A)$.

In general, the function “ $Prob$ ” is not strictly defined in the mathematical sense and we can use it more informal. A formal concept is the function “ P ”, a probability measure.

Probability Distribution: Informal Definition

Definition (informal):

Let $A \subset \Omega$. Instead of $Prob(D \in A)$ we write $P(A)$ and we call P a probability distribution or probability measure if P meets the following criteria:

$$P(\Omega) = 1$$

$$P(\emptyset) = 0$$

$$P(A) \geq 0 \text{ for any } A \subset \Omega$$

If $A \subset \Omega$ and $B \subset \Omega$ then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Special case: if $A \cap B = \emptyset \Rightarrow P(A \cap B) = 0$ and thus $P(A \cup B) = P(A) + P(B)$.

Probability Distribution: Formal Definition

Definition (formal):

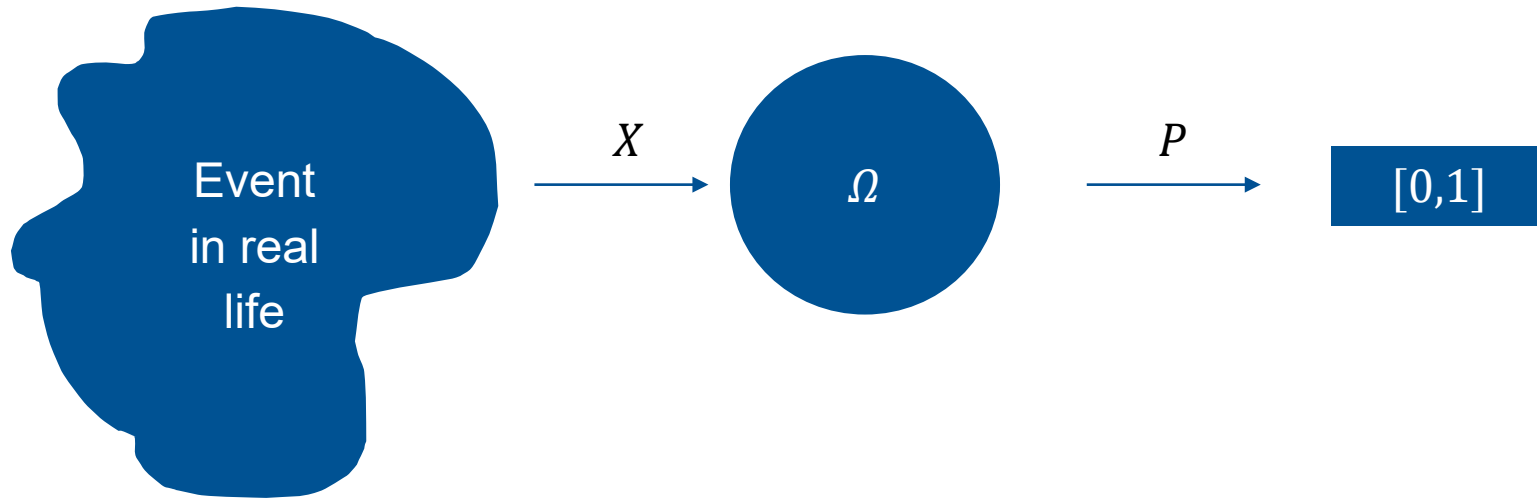
Let Ω be the sample space of a random variable with σ -algebra Σ , that is, Σ comprises all measurable subsets of Ω . A map $P: \Sigma \rightarrow [0,1]$ is called probability distribution or probability measure if it satisfies

$$P(\Omega) = 1$$

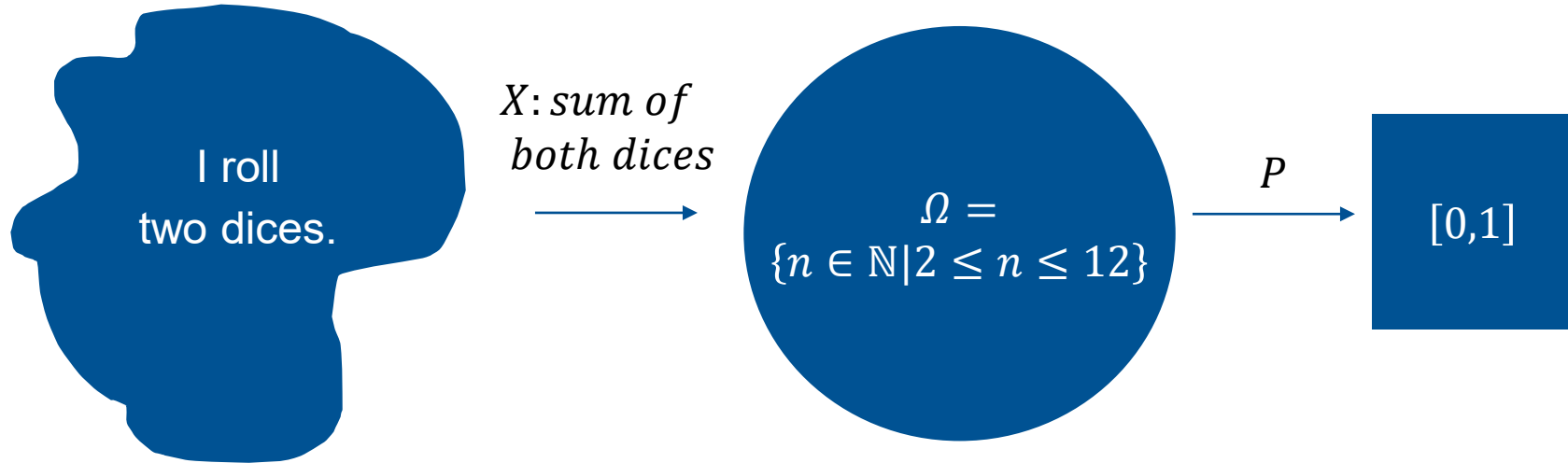
And for each sequence of pairwise disjoint sets $A_1, A_2, A_3, \dots$ in Σ it is:

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

Modeling Real Life



Example: Rolling Dices



X	2	3	4	5	6	7	8	9	10	11	12
$P(X)$	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36
$P(X)$ (in %)	2.78%	5.56%	8.33%	11.11%	13.89%	16.67%	13.89%	11.11%	8.33%	5.56%	2.78%

Discrete Random Variables

Discrete versus Continuous

Discrete means countable. A discrete variable can still refer to an infinite amount. For instance, natural numbers are discrete yet infinite. You can count them like 1, 2, 3, ...

Continuous means uncountable. For instance, real numbers are uncountable. You cannot count them. For instance, which real number would be the number next to 1?

Discrete versus Continuous

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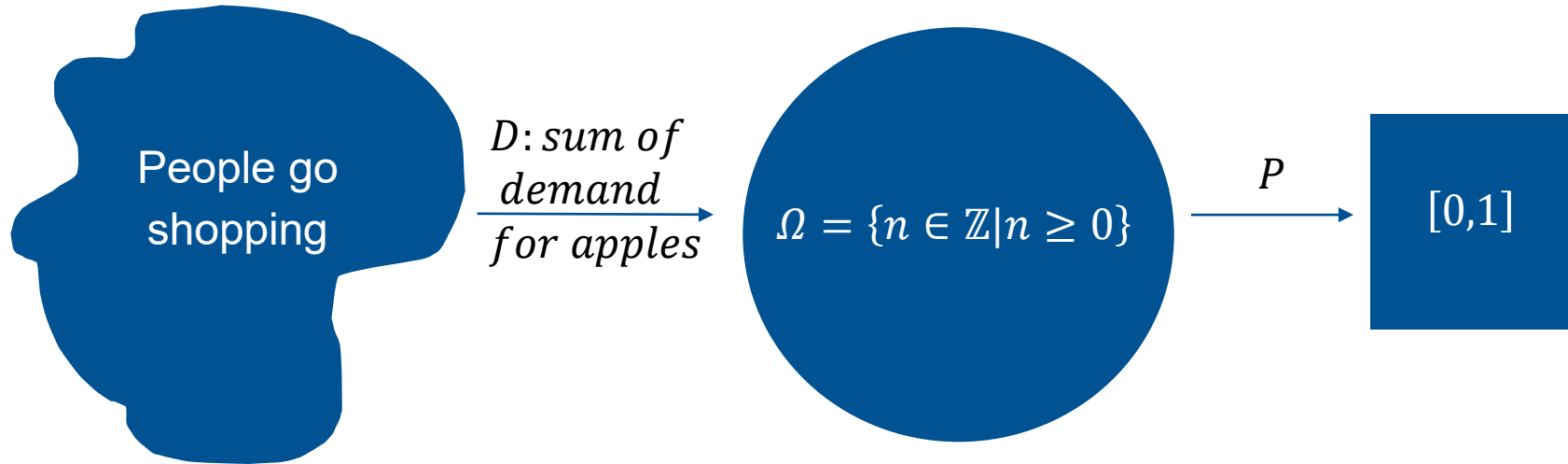
Continuous means uncountable. For instance, real numbers are uncountable. You cannot count them. For instance, which real number would be the number next to 1?

Definition

- We use $\mathbb{N}$ to refer to natural numbers (1, 2, 3, ...),
- we use $\mathbb{Z}$ to refer to integers (... , -3, -2, -1, 0, 1, 2, 3, ...),
- we use $\mathbb{R}$ to refer to real numbers,
- we use $\mathbb{R}^+$ to refer to positive real numbers excluding 0, and
- we use $\mathbb{R}_0^+$ to refer to positive real numbers including 0.

Discrete Demand

E.g., consider the introductory example for demand for apples.



Discrete Demand and Probability

Usually, in operations, we are interested in the question: How likely will all demand be satisfied?

Let $\Omega = \{n \in \mathbb{Z} | n \geq 0\}$ be the domain of random variable D representing demand. Define $A_n := \{\omega \in \Omega | \omega \leq n\}$. Then A_n captures all demand realizations such that the demand does not exceed n .

Put differently, if you order n units before demand is realized, the probability to satisfy demand is $P(A_n)$.

Note, for discrete random variables it holds that $A_n = \{0\} \cup \{1\} \cup \{2\} \cup \dots \cup \{n\}$.

Hence $P(A_n) = P(\{0\}) + P(\{1\}) + P(\{2\}) + \dots + P(\{n\})$.

Discrete Demand and Probability

Question: How do we calculate $P(\{k\})$?

That is, what is a suitable distribution to model discrete demand?

Suitable Discrete Demand Distributions

Bernoulli distribution

Discrete uniform distribution

Poisson distribution

Poisson Distribution

Poisson Distribution (Motivation)

Assume you are interested in the number of customers that arrive at a store within a fixed amount of time (e.g., one hour, one day, or one week).

Assume customers appear independently of other customers.

This is a typically situation leading to the use of the Poisson distribution.

Poisson Distribution (Motivation)

Definition:

Let D be a random variable with sample space $\Omega = \{n \in \mathbb{Z} | n \geq 0\}$ and probability measure $P(\{k\}) = \frac{\lambda^k e^{-\lambda}}{k!}$.
Then the distribution of D is called Poisson distribution with parameter $\lambda \in \mathbb{R}^+$.

Poisson Distribution (Formula)

$$P(\{k\}) =: f(k; \lambda) = \frac{\lambda^k e^{-\lambda}}{k!}$$

$e = 2.71828 \dots$ *Euler's number*

"!" = *factorial, that is, $n! := 1 \cdot 2 \cdot 3 \cdot \dots \cdot n$*

Poisson Distribution Calculation

$$P(\{k\}) =: f(k; \lambda) = \frac{\lambda^k e^{-\lambda}}{k!}$$

Example $k = 0$:

$$f(0; \lambda) = \frac{\lambda^0 e^{-\lambda}}{0!} = e^{-\lambda}$$

Example $k = 1$:

$$f(1; \lambda) = \frac{\lambda^1 e^{-\lambda}}{1!} = \lambda e^{-\lambda}$$

Example $k = 2$:

$$f(2; \lambda) = \frac{\lambda^2 e^{-\lambda}}{2!} = 0.5 \lambda^2 e^{-\lambda}$$

Is the Poisson Distribution a Distribution?

$$P(\{k\}) > 0?$$

Is the Poisson Distribution a Distribution?

$$P(\{k\}) \geq 0?$$
$$\frac{\lambda^k e^{-\lambda}}{k!} > 0$$

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$$P(\Omega) = 1?$$

Is the Poisson Distribution a Distribution?

$$P(\Omega) = 1?$$

$$P(\Omega) = P(\{0\}) + P(\{1\}) + P(\{2\}) + \dots$$

$$\begin{aligned} &= \sum_{k=0}^{\infty} P(\{k\}) = \sum_{k=0}^{\infty} \left(\frac{\lambda^k e^{-\lambda}}{k!} \right) = e^{-\lambda} \underbrace{\sum_{k=0}^{\infty} \left(\frac{\lambda^k}{k!} \right)}_{\text{= Taylor series of } e^x \text{ at } x = \lambda.} = e^{-\lambda} e^{\lambda} = 1 \end{aligned}$$

Poisson Probability Distribution – Interpreting λ

How can we interpret λ ?

Imagine, we could observe many realizations of random variable D , then the probability times the number of realizations can be interpreted as the frequency, that is, how often, on average, each realization occurs.

Hence, after N repetitions we expect to observe the following frequencies:

k	Frequency
0	$Ne^{-\lambda}$
1	$N\lambda e^{-\lambda}$
2	$N0.5\lambda^2 e^{-\lambda}$
...	...
k	$N \frac{\lambda^k e^{-\lambda}}{k!}$
...	...

Excursus: How do you calculate the average?

Assume you are given the numbers 2, 3, 3, 4, 4, 4, 8 and want to calculate the average, A , then

$$A = \frac{2 + 3 + 3 + 4 + 4 + 4 + 8}{7} = \frac{28}{7} = 4$$

Alternatively, you could write

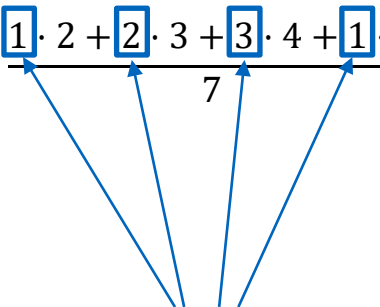
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Alternatively, you could write

$$A = \frac{\boxed{1} \cdot 2 + \boxed{2} \cdot 3 + \boxed{3} \cdot 4 + \boxed{1} \cdot 8}{7} = \frac{28}{7} = 4$$


That is, you can multiply each number with its **frequency** and take the sum.

Back to λ , let's calculate the average!

k	Frequency
0	$Ne^{-\lambda}$
1	$N\lambda e^{-\lambda}$
2	$N0.5\lambda^2 e^{-\lambda}$
...	...
k	$N \frac{\lambda^k e^{-\lambda}}{k!}$
...	...

$$A = \frac{Ne^{-\lambda}0 + N\lambda e^{-\lambda}1 + N0.5\lambda^2 e^{-\lambda}2 + \dots}{N}$$

$$= \sum_{k=0}^{\infty} \left(\frac{Nk\lambda^k e^{-\lambda}}{k! N} \right) = \sum_{k=0}^{\infty} \left(\frac{k\lambda^k e^{-\lambda}}{k!} \right)$$

Simplify the Average

$$\begin{aligned} A &= \sum_{k=0}^{\infty} \left(\frac{k \lambda^k e^{-\lambda}}{k!} \right) = \frac{0 \lambda^0 e^{-\lambda}}{0!} + \sum_{k=1}^{\infty} \left(\frac{k \lambda^k e^{-\lambda}}{k!} \right) = 0 + \sum_{k=0}^{\infty} \left(\frac{(k+1) \lambda^{(k+1)} e^{-\lambda}}{(k+1)!} \right) \\ &= \sum_{k=0}^{\infty} \left(\frac{\lambda \lambda^k e^{-\lambda}}{k!} \right) = \lambda \sum_{k=0}^{\infty} \left(\frac{\lambda^k e^{-\lambda}}{k!} \right) = \lambda \end{aligned}$$

That is, if we were to repeat realizations of the random variable N times, the average would be λ . We call λ the **expected value** and write $\lambda = \mathbb{E}[D]$.

Poisson Probability Distribution – Calculations

Calculating with the Poisson Distribution

Assume demand D is a random variable which follows a Poisson Distribution with expected value λ . For instance, assume $\lambda = 3$. How do you calculate $P(A_5)$, that is, the probability that demand does not exceed 5?

Calculating with the Poisson Distribution

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Recall that for discrete random variables: $P(A_n) = P(\{0\}) + P(\{1\}) + P(\{2\}) + \dots + P(\{n\})$

Therefore, $P(\{0,1,2,3,4,5\}) = P(\{0\}) + P(\{1\}) + P(\{2\}) + P(\{3\}) + P(\{4\}) + P(\{5\})$

With $\lambda = 3$,
$$f(k; 3) = \frac{3^k e^{-3}}{k!}$$

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With $\lambda = 3$,
$$f(k; 3) = \frac{3^k e^{-3}}{k!}$$

$$P(\{0\}) = e^{-3}$$

$$P(\{1\}) = 3e^{-3}$$

$$P(\{2\}) = 4.5e^{-3}$$

$$P(\{3\}) = 4.5e^{-3}$$

$$P(\{4\}) = 3.375e^{-3}$$

$$P(\{5\}) = 2.025e^{-3}$$

$$\left. \begin{array}{l} P(\{0\}) = e^{-3} \\ P(\{1\}) = 3e^{-3} \\ P(\{2\}) = 4.5e^{-3} \\ P(\{3\}) = 4.5e^{-3} \\ P(\{4\}) = 3.375e^{-3} \\ P(\{5\}) = 2.025e^{-3} \end{array} \right\} P(A_5) = 18.4e^{-3} \approx 0.916$$

The Same but Improved Structure

k	$P(\{k\})$	$P(A_k)$
0	0.05	0.05
1	0.15	0.20
2	0.22	0.42
3	0.22	0.65
4	0.17	0.82
5	0.10	0.92

Tells you for each k how likely the demand will not exceed this level.

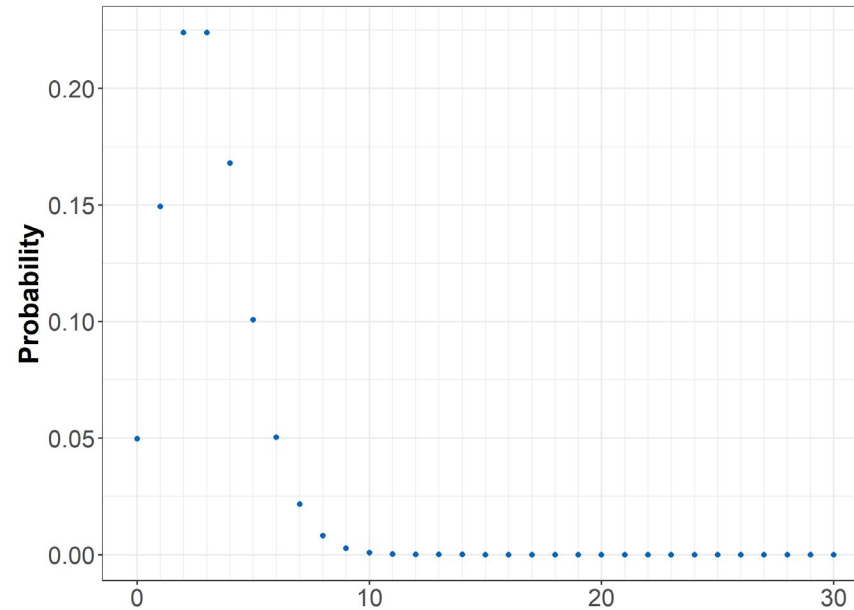
$$P(A_2) = P(A_1) + P(\{2\})$$

Is there a simple formula?

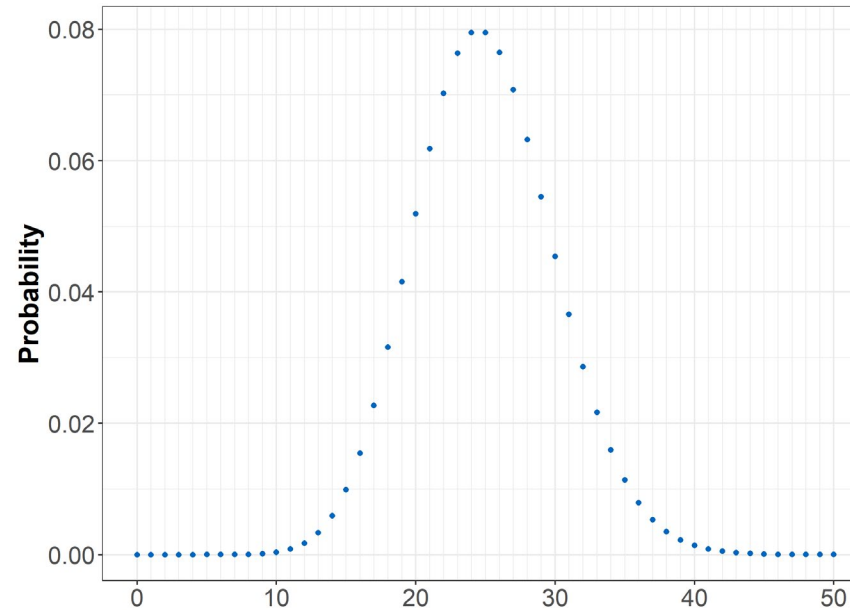
$$P(A_n) = \sum_{k=0}^n \left(\frac{\lambda^k e^{-\lambda}}{k!} \right)$$

Poisson Probability Distribution – Graphical Interpretation

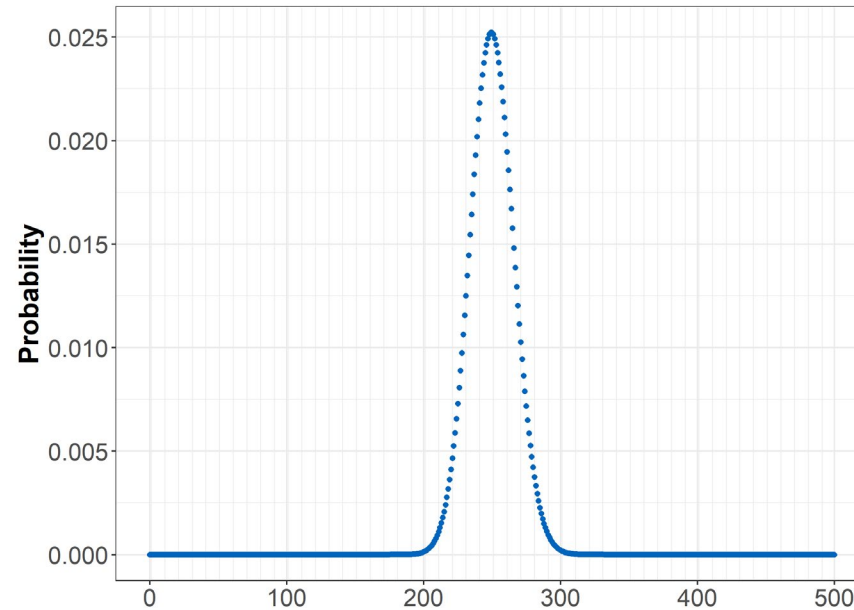
Poisson Distribution with $\lambda = 3$



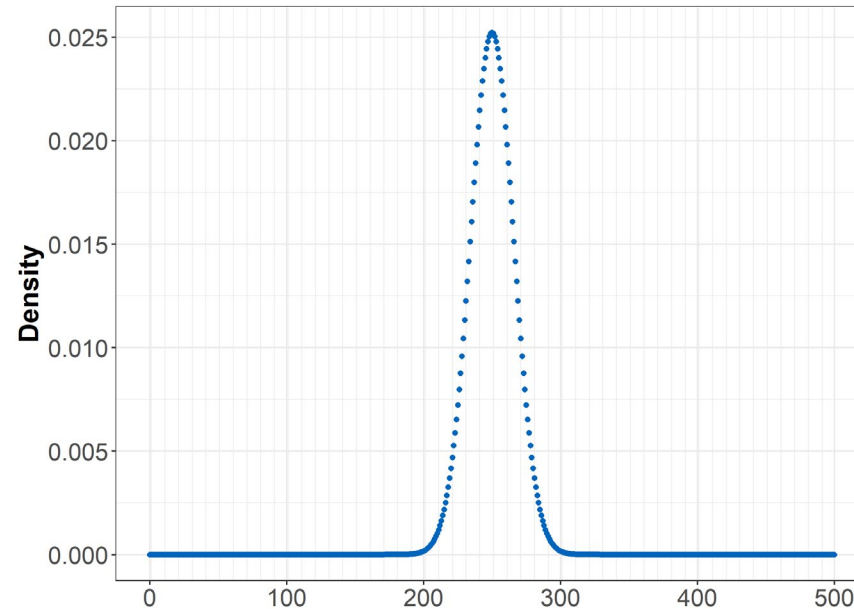
Poisson Distribution with $\lambda = 25$



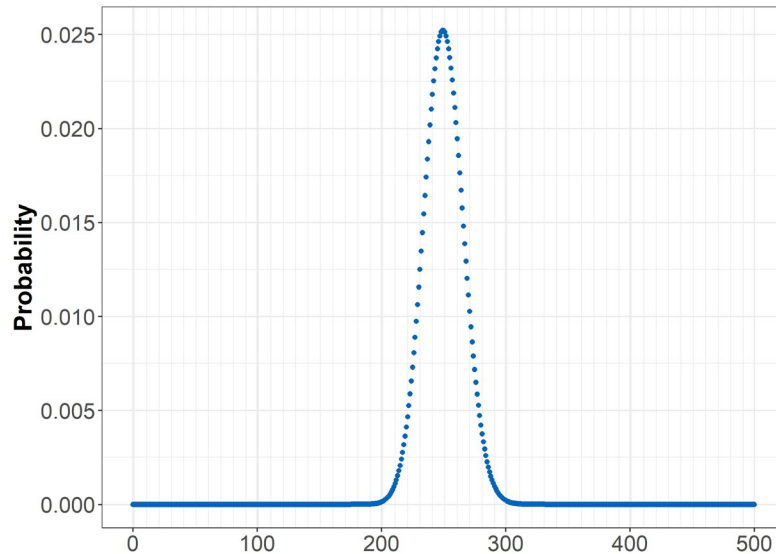
Poisson Distribution with $\lambda = 250$



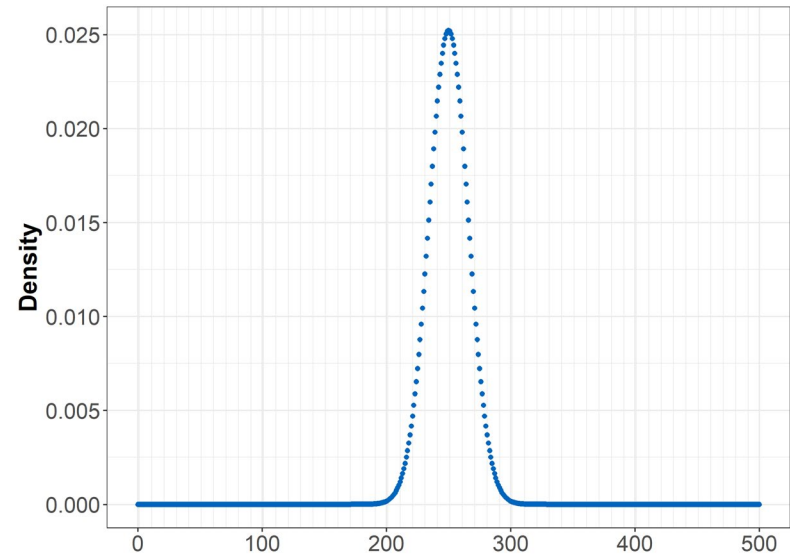
Normal Distribution with $\mu = 250, \sigma^2 = 250$



Comparison: Poisson versus Normal Random Variable



Poisson Distribution with $\lambda = 250$



Normal Distribution with $\mu = 250, \sigma^2 = 250$

Continuous Random Variables

Continuous Random Variables

If we consider X to be a continuous random variable, we usually consider a sample space $\Omega \subset \mathbb{R}$ and usually Ω is some interval.

Example: Consider the following random experiment. Take a bottle with exactly 1 liter of water. Open it, turn it around for a few seconds. Turn it back. How much water is left?

Continuous Random Variables

If we consider X to be a continuous random variable, we usually consider a sample space $\Omega \subset \mathbb{R}$ and usually Ω is some interval.

Example: Consider the following random experiment. Take a bottle with exactly 1 liter of water. Open it, turn it around for a few seconds. Turn it back. How much water is left?

We could use continuous random variable X with support $[0,1]$ to express the remaining amount of water. P could be the probability that a given amount within interval A is left. For instance, we could be interested in $P([0.2,0.4])$.

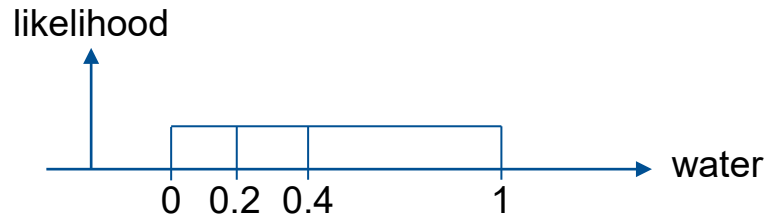
Challenges Pertaining to Continuous Random Variables

How to sum up probabilities?

What is the probability of one specific outcome? E.g., exactly 0.25 liters are left, that is, $P(\{0.25\})$?

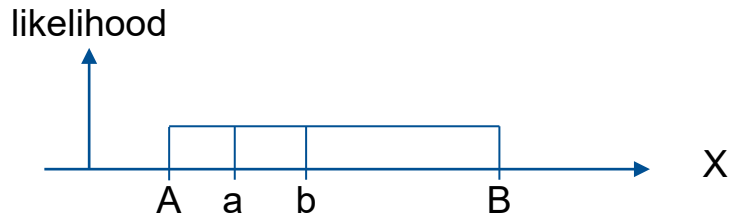
Uniform Distribution

Continuous Uniform Distribution



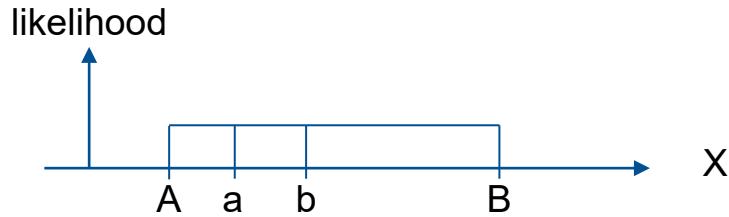
$$P([0,1]) = 1 \text{ and } P([0.2,0.4]) = \frac{0.4-0.2}{1} = 0.2$$

Continuous Uniform Distribution



$P([A, B]) = 1$ and $P([a, b]) = \frac{b-a}{B-A}$ for any uniform distribution with $A \leq a \leq b \leq B$.

Continuous Uniform Distribution



$P([A, B]) = 1$ and $P([a, b]) = \frac{b-a}{B-A}$ for any uniform distribution with $A \leq a \leq b \leq B$.

Let $A \leq a \leq B$, then $P(\{a\}) = P([a, a]) = \frac{a-a}{B-A} = 0$ (!)

Continuous Random Variables (Theory)

Cumulative Distribution Function (CDF)

Let $\Omega \subset \mathbb{R}$ be the sample space of random variable X and let Ω be an interval. We are often interested in intervals in the form of $[\inf(\Omega), a]$. (Here, the function $\inf$ is the infimum function, which gives the highest lower bound of set. If this set is closed, $\inf = \min$.) If, for instance, $\Omega = \mathbb{R}$, $[\inf(\Omega), a] = [-\infty, a]$; if, instead, $\Omega = \mathbb{R}_0^+$, then $[\inf(\Omega), a] = [0, a]$.

We are then interested in $P([-\infty, a])$ or $P([0, a])$, respectively. That is, the probability that the realization of X is below or equal to a .

Definition:

Let interval $\Omega \subset \mathbb{R}$ be the sample space of random variable X with probability measure P .

Then we define $F(a) := P([\inf(\Omega), a])$ as the cumulative distribution function (CDF) of X under probability measure P . In short, we say X has CDF F .

If $\Omega = \mathbb{R}$, $F(a) = P([-\infty, a])$. If $\Omega = \mathbb{R}_0^+$, $F(a) = P([0, a])$.

Cumulative Distribution Function (CDF)

Proposition:

Let interval $\Omega \subset \mathbb{R}$ be the sample space of a continuous random variable X with continuous probability measure P . Then $P([a, a]) = 0$.

Calculating with Cumulative Distribution Function (CDF)

Let $\Omega \subset \mathbb{R}$ be the sample space of random variable X with probability measure P and cumulative distribution function $F(a)$. Let $a, b \in \Omega$ and $a \leq b$. Then $[-\infty, a] \cup [a, b] = [-\infty, b]$. Further, $[-\infty, a] \cap [a, b] = [a, a]$. Hence,

$$\begin{aligned} P([-\infty, b]) &= P([a, b] \cup [-\infty, a]) \\ &= P([a, b]) + P([-\infty, a]) - P([a, b] \cap [-\infty, a]) \\ &= P([a, b]) + P([-\infty, a]) - P([a, a]) \\ &= P([a, b]) + P([-\infty, a]) \end{aligned}$$

Therefore,

$$\begin{aligned} P([-\infty, b]) - P([-\infty, a]) &= P([a, b]) \\ \Rightarrow P([a, b]) &= F(b) - F(a) \end{aligned}$$

CDF: Uniform Distribution Example

Let $\Omega = [A, B]$, $\inf(\Omega) = A$, with cumulative distribution function (CDF) $F(a) = P([A, a])$,

$$F(a) = \begin{cases} 0 & \text{if } a \leq A \\ \frac{a-A}{B-A} & \text{if } A \leq a \leq B \\ 1 & \text{if } a > B \end{cases}$$

Probability Density Function (PDF)

Definition:

Let X be a random variable with sample space Ω and cumulative distribution function F . The first derivative of F is called Probability Density Function (PDF) and we write $f(a) := \frac{d}{da}F(a)$.

According to the fundamental theorem of calculus it holds that

$$F(A) = \int_A f(x)dx .$$

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Examples:

If $\Omega = \mathbb{R}$, $F(a) = \int_{-\infty}^a f(x)dx$.

If $\Omega = \mathbb{R}_0^+$, $F(a) = \int_0^a f(x)dx$

PDF: Uniform Distribution Example

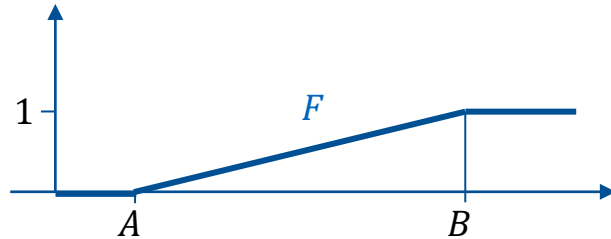
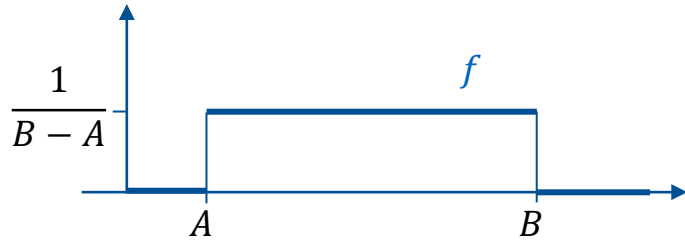
$$f(a) = \begin{cases} 0 & \text{if } a \leq A \\ \frac{1}{B-A} & \text{if } A \leq a \leq B \\ 0 & \text{if } a > B \end{cases}$$

CDF and PDF: Important Properties

Let X be a random variable with sample space Ω , cumulative distribution function F and probability density function f . Then

- 1) $f(a) \geq 0$,
- 2) $F(a) \geq 0$,
- 3) $F(a) \leq 1$,
- 4) $F(a)$ increases (weakly) monotonically, and
- 5) $f(a)$ may exceed 1.

CDF and PDF: Uniform Distribution Example



Expected Value

Expected Value of Continuous Random Variable

Definition:

Let X be a random variable with sample space Ω , cumulative distribution function F and probability density function f . If $\int_{\Omega} f(x)xdx < \infty$ we call $\int_{\Omega} f(x)xdx$ the expected value of X and write

$$\mathbb{E}[X] = \int_{\Omega} f(x)xdx$$

and if Ω is an interval we can write

$$\mathbb{E}[X] = \int_{\inf \Omega}^{\sup \Omega} f(x)xdx .$$

(Again, recall that $\inf$ is the greatest lower bound of a set and $\sup$ is the smallest upper bound of a set and for closed sets these equal $\min$ and $\max$, respectively.)

Interpretation of Expected Value

Consider the sequence of random variables $X_1, X_2, X_3, \dots$ which all have the same support Ω and all have the same probability density function $f(a)$. Let $x_1, x_2, x_3, \dots$ denote the series of realizations of these random variables. Then,

1. $\mathbb{E}[X_1] = \mathbb{E}[X_2] = \mathbb{E}[X_3] = \dots$ (all variables have the same expected value)
2. $\mathbb{E}[X_1] = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n x_i$ (the average of realizations of the same random variable converges to the expected value)

Example: Expected Value of Uniform Distribution

Let random variable X be uniformly distributed on the interval $[A, B]$. That is, $\Omega = [A, B]$ and

$$f(a) = \begin{cases} 0 & \text{if } a \leq A \\ \frac{1}{B-A} & \text{if } A \leq a \leq B \\ 0 & \text{if } a > B \end{cases}$$

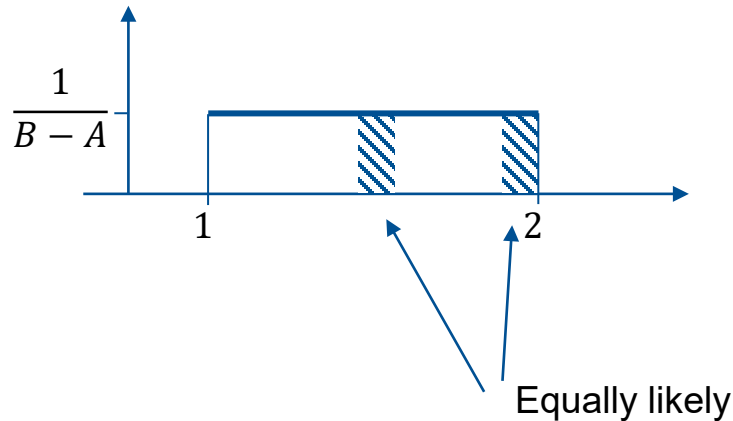
Then

$$\mathbb{E}[X] = \int_{\inf \Omega}^{\sup \Omega} f(x)x dx = \int_A^B \frac{1}{B-A} x dx = \frac{1}{B-A} \int_A^B x dx = \frac{1}{B-A} \frac{1}{2} (B^2 - A^2) = \frac{1}{2} \frac{(B-A)(B+A)}{B-A} = \frac{1}{2} (B+A)$$

Normal Distribution

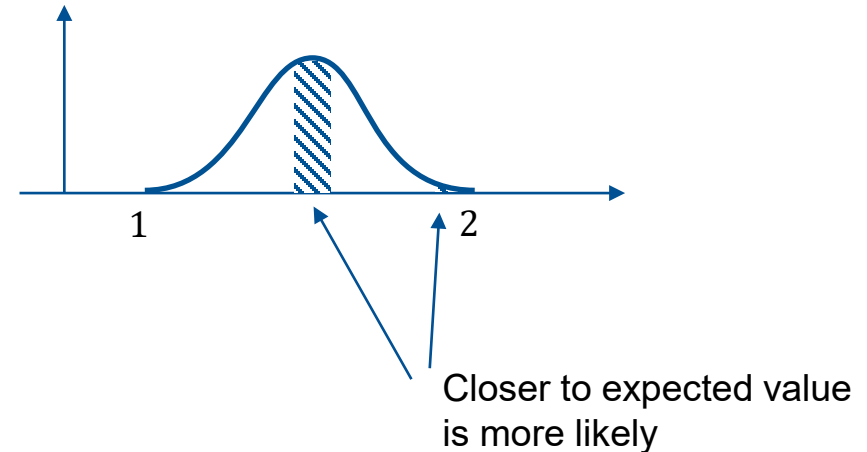
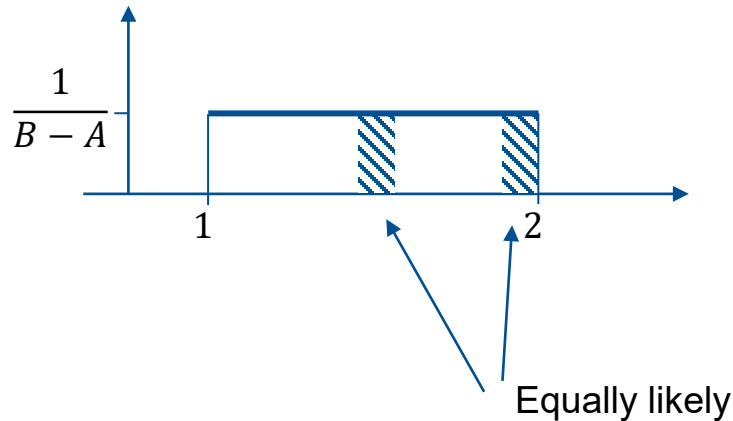
Plausibility of Uniform Distribution

Let X be a random variable with sample space $[1,2]$ and uniform distribution. Then the expected value is 1.5; further, $P([1.45,1.55]) = 0.1 = P([1.9,2.0])$.



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Normal Distribution

Definition:

Let X be a random variable with sample space $\Omega = \mathbb{R}$ and probability density function (PDF) f .

If $f(a; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\left(\frac{(a-\mu)^2}{2\sigma^2}\right)}$ we say that X is normally distributed or follows a normal distribution with mean μ and standard deviation σ . To be explicit, we sometimes write $f_{\mathcal{N}}(a; \mu, \sigma)$.

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If $\mu = 0$ and $\sigma = 1$, that is, $f(a; 0, 1) = \frac{1}{\sqrt{2\pi}} e^{-\left(\frac{a^2}{2}\right)}$ we say the X is **standard normally distributed** or **follows a standard normal distribution**. We sometimes write $f_{0,1}(a)$.

Normal Distribution

Proposition:

Let X be a normally distributed random variable with mean μ and standard deviation σ . The corresponding CDF,

$$F_{\mathcal{N}}(a; \mu, \sigma) = \int_{-\infty}^a f_{\mathcal{N}}(x; \mu, \sigma) dx = \int_{-\infty}^a \frac{1}{\sqrt{2\pi}\sigma} e^{-\left(\frac{(x-\mu)^2}{2\sigma^2}\right)} dx ,$$

cannot be expressed in closed form and only be evaluated numerically.

Working with the Normal Distribution

Excel

→ `NORM.DIST(x,mean,standard_dev,cumulative)`

Other software packages (e.g., R)

→ `pnorm(q, mean = 0, sd = 1)`

z-tables



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0003	.0003	.0003	.0003	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0006	.0005	.0005
-3.1	.0010	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0031	.0030	.0029	.0028	.0027
-2.6	.0047	.0045	.0044	.0043	.0041	.0041	.0040	.0039	.0038	.0037
-2.5	.0062	.0060	.0059	.0057	.0056	.0054	.0053	.0052	.0051	.0050
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0440	.0430	.0422	.0413	.0405	.0397	.0389	.0381	.0374	.0367
-1.6	.0540	.0527	.0520	.0510	.0502	.0493	.0485	.0475	.0468	.0461
-1.5	.0660	.0645	.0641	.0630	.0619	.0609	.0599	.0590	.0581	.0573
-1.4	.0808	.0793	.0779	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0985	.0969	.0954	.0938	.0921	.0905	.0889	.0873	.0858	.0843
-1.2	.1191	.1173	.1157	.1139	.1121	.1103	.1085	.1068	.1051	.1034
-1.1	.1467	.1446	.1425	.1403	.1381	.1359	.1336	.1314	.1291	.1269
-1.0	.1867	.1841	.1814	.1786	.1759	.1731	.1703	.1675	.1647	.1619
-0.9	.2398	.2368	.2338	.2308	.2276	.2245	.2213	.2181	.2149	.2116
-0.8	.2967	.2935	.2902	.2869	.2835	.2801	.2767	.2732	.2697	.2661
-0.7	.3643	.3609	.3574	.3538	.3501	.3464	.3427	.3389	.3351	.3313
-0.6	.4427	.4389	.4350	.4311	.4271	.4231	.4190	.4149	.4107	.4065
-0.5	.5407	.5367	.5326	.5284	.5241	.5198	.5155	.5112	.5068	.5024
-0.4	.6406	.6364	.6321	.6277	.6232	.6187	.6141	.6095	.6049	.6002
-0.3	.7420	.7374	.7327	.7279	.7231	.7182	.7133	.7083	.7033	.6982
-0.2	.8438	.8389	.8339	.8288	.8236	.8183	.8129	.8075	.8021	.7966
-0.1	.9452	.9401	.9348	.9293	.9237	.9180	.9122	.9063	.9003	.8942
0.0	1.0000	.9980	.9960	.9940	.9920	.9900	.9880	.9860	.9840	.9820

Normal Distribution and z-table

z-tables and Standard Normal Distributed Random Variables

Let X be a **standard** normally distributed random variable with CDF F . Then $P(X \leq a)$, that is, $F_{0,1}(a)$, can be obtained from so called z-tables.

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Let X be a **standard** normally distributed random variable with CDF F . Then $P(X \leq a)$, that is, $F_{0,1}(a)$ can be obtained from so called z-tables.

Example:

$$a = -2.74$$

z	.00	.01	.02	.03	.04	.05	.06	.0
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.00
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.00
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.00
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.00
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.00
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.00
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.00
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.00
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.00
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.00
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.00
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.00
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.01
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.01
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.01
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.02
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.03
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.03
-1.6	.0540	.0529	.0519	.0509	.0500	.0490	.0480	.04

z-tables and Standard Normal Distributed Random Variables

Let X be a **standard** normally distributed random variable with CDF F . Then $P(X \leq a)$, that is, $F_{0,1}(a)$ can be obtained from so called z-tables.

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z	.00	.01	.02	.03	.04	.05	.06	.0
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.00
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.00
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.00
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.00
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.00
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.00
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.00
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.00
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.00
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.00
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.00
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-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.01
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.01
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.02
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.03
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.03
-1.6	.0540	.0530	.0520	.0510	.0500	.0490	.0480	.04

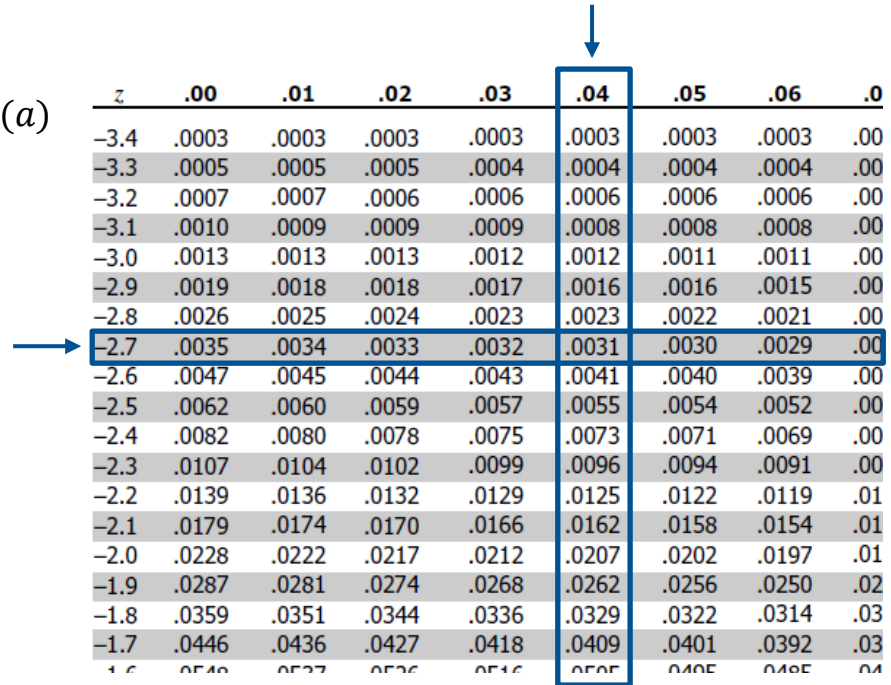
z-tables and Standard Normal Distributed Random Variables

Let X be a **standard** normally distributed random variable with CDF F . Then $P(X \leq a)$, that is, $F_{0,1}(a)$ can be obtained from so called z-tables.

Example:

$$a = -2.74$$

$$F_{0,1}(a) = 0.0031$$

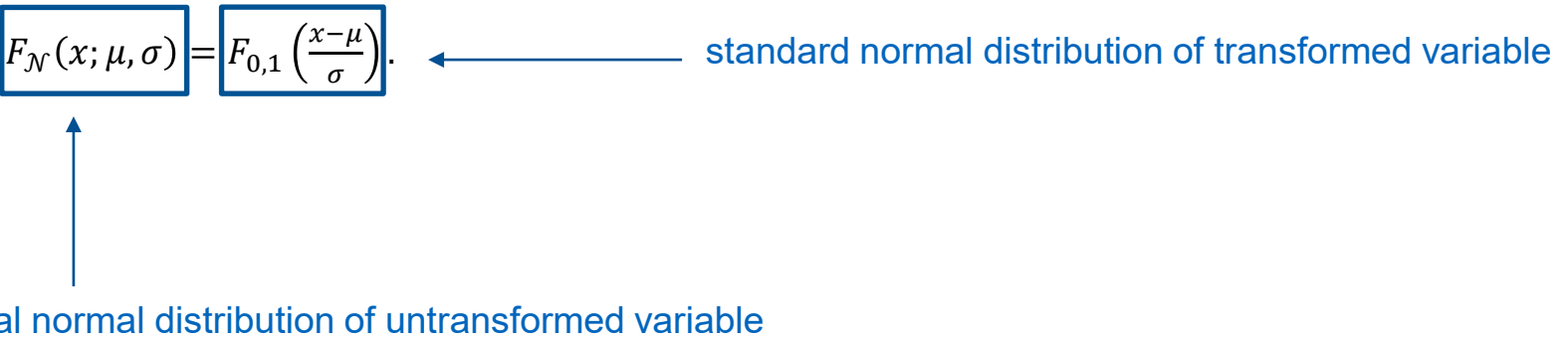


z	.00	.01	.02	.03	.04	.05	.06	.0
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.00
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.00
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.00
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.00
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.00
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.00
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.00
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.00
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.00
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.00
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.00
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.00
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.01
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.01
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.01
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.02
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.03
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.03
-1.6	.0540	.0537	.0534	.0531	.0528	.0525	.0522	.04

z-tables and General Normal Distributed Random Variables

Proposition:

Let X be a normally distributed random variable with CDF $F_{\mathcal{N}}$ and mean μ and standard deviation σ . Then there exists a random variable Z that is standard normally distributed and $Z = \frac{X - \mu}{\sigma}$.

It holds that $F_{\mathcal{N}}(x; \mu, \sigma) = F_{0,1}\left(\frac{x - \mu}{\sigma}\right)$. 

general normal distribution of untransformed variable

standard normal distribution of transformed variable

z-tables and Normal Distributed Random Variables

Example:

Let X be a normally distributed random variable with CDF F and mean $\mu = 2$ and standard deviation $\sigma = 6$. What is $F_{\mathcal{N}}(4)$?

z-tables and Normal Distributed Random Variables

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Let X be a normally distributed random variable with CDF F and mean $\mu = 2$ and standard deviation $\sigma = 6$. What is $F_{\mathcal{N}}(4)$?

Step 1 (normalize):

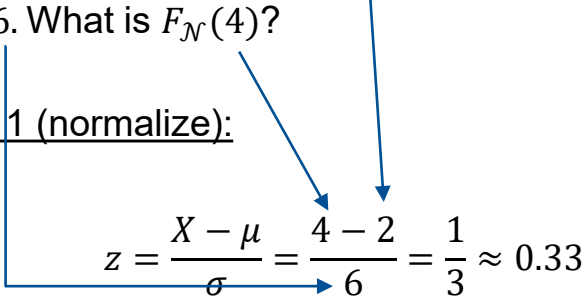
$$z = \frac{X - \mu}{\sigma}$$

z-tables and Normal Distributed Random Variables

Example:

Let X be a normally distributed random variable with CDF F and mean $\mu = 2$ and standard deviation $\sigma = 6$. What is $F_{\mathcal{N}}(4)$?

Step 1 (normalize):



The diagram shows the formula $z = \frac{X - \mu}{\sigma} = \frac{4 - 2}{6} = \frac{1}{3} \approx 0.33$. Three blue arrows originate from the text above: one points from '4' in $F_{\mathcal{N}}(4)$ to the '4' in the numerator, one points from ' $\mu = 2$ ' to the '2' in the numerator, and one points from ' $\sigma = 6$ ' to the '6' in the denominator.

$$z = \frac{X - \mu}{\sigma} = \frac{4 - 2}{6} = \frac{1}{3} \approx 0.33$$

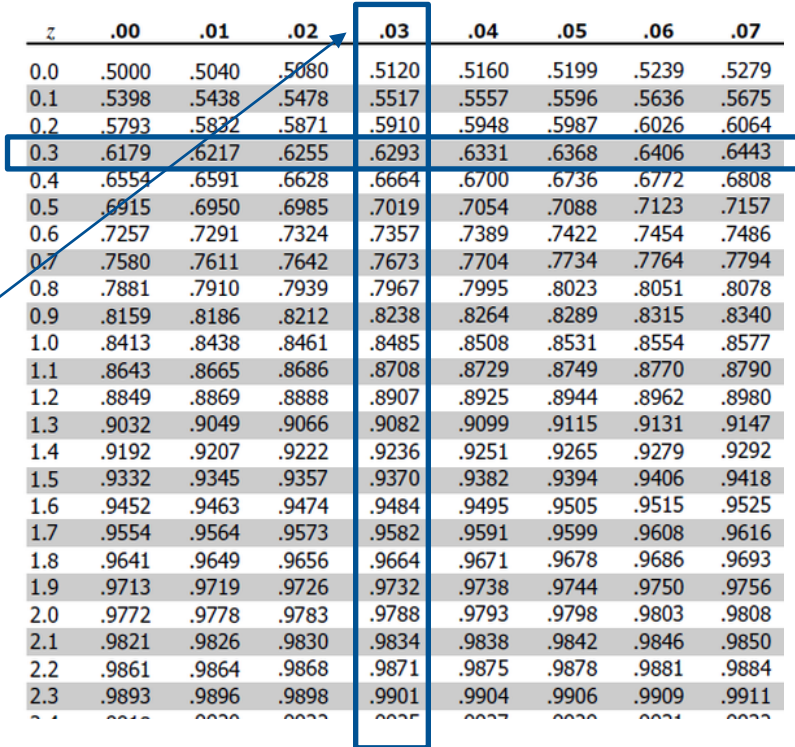
z-tables and Normal Distributed Random Variables

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Let X be a normally distributed random variable with CDF F and mean $\mu = 2$ and standard deviation $\sigma = 6$. What is $F_{\mathcal{N}}(4)$?

Step 1 (normalize):

$$z = \frac{X - \mu}{\sigma} = \frac{4 - 2}{6} = \frac{1}{3} \approx 0.33$$



z	.00	.01	.02	.03	.04	.05	.06	.07
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911

z-tables and Normal Distributed Random Variables

Example:

Let X be a normally distributed random variable with CDF F and mean $\mu = 2$ and standard deviation $\sigma = 6$. What is $F_{\mathcal{N}}(4)$?

Step 1 (normalize):

$$z = \frac{X - \mu}{\sigma} = \frac{4 - 2}{6} = \frac{1}{3} \approx 0.33$$

Step 2 (use z-table):

$$F_{\mathcal{N}}(4) = F_{0,1}\left(\frac{1}{3}\right) = 0.6293$$

z	.00	.01	.02	.03	.04	.05	.06	.07
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911

Insight through the z-table

$$F_{0,1}(0) = 0.5$$

$$x > 0 \Leftrightarrow F_{0,1}(x) > 0.5$$

$$x < 0 \Leftrightarrow F_{0,1}(x) < 0.5$$

$$F_{0,1}(-1.96) = 0.025$$

$$F_{0,1}(1.96) = 0.975$$

If X is standard normally distributed, $P([-1.96, 1.96]) = 0.95$.

If X is normally distributed with mean μ and standard deviation σ , then $P([\mu - 1.96\sigma, \mu + 1.96\sigma]) = 0.95$.

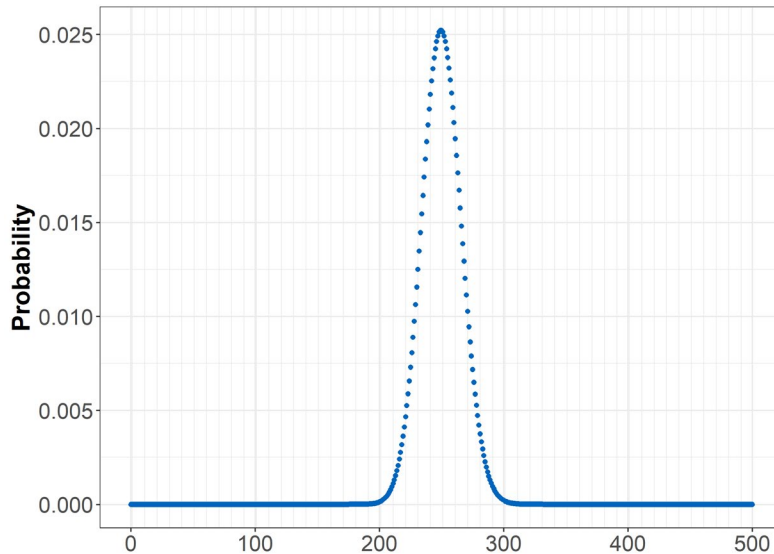
Further, $F_{0,1}(2.32) \approx 0.99$; that is, if quantity $a > \mu + 2.32\sigma$ is ordered all demand will be met with probability of 99%.

Modeling Uncertain Demand

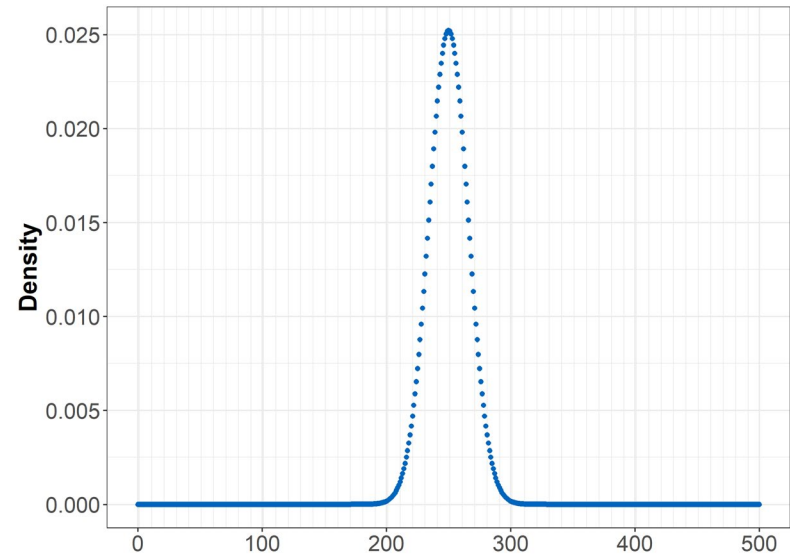
Poisson Distribution versus Normal Distribution

Poisson Distribution	Normal Distribution
Discrete distribution	Continuous distribution
One parameter (λ)	Two parameters (μ, σ)
Iterative evaluation	Explicit evaluation

Comparison Poisson versus Normal Random Variable

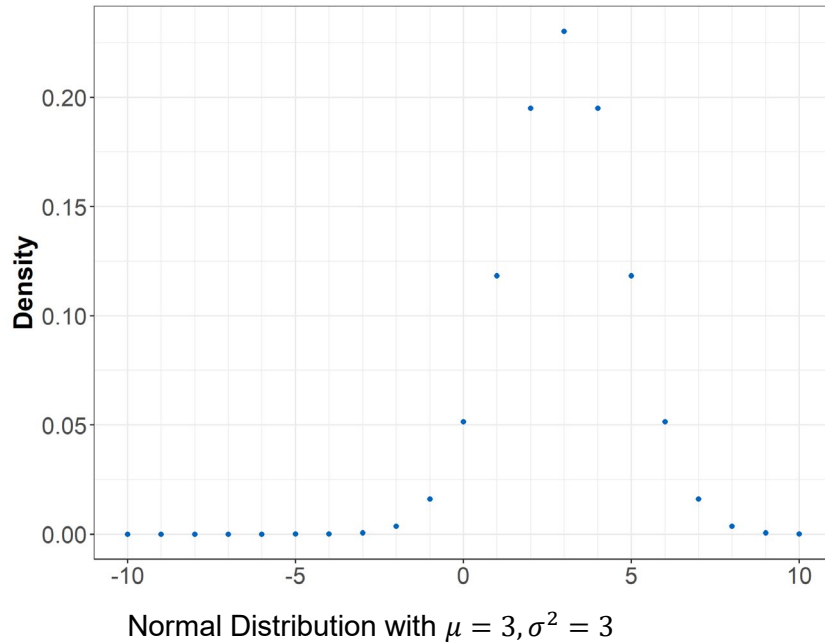


Poisson Distribution with $\lambda = 250$

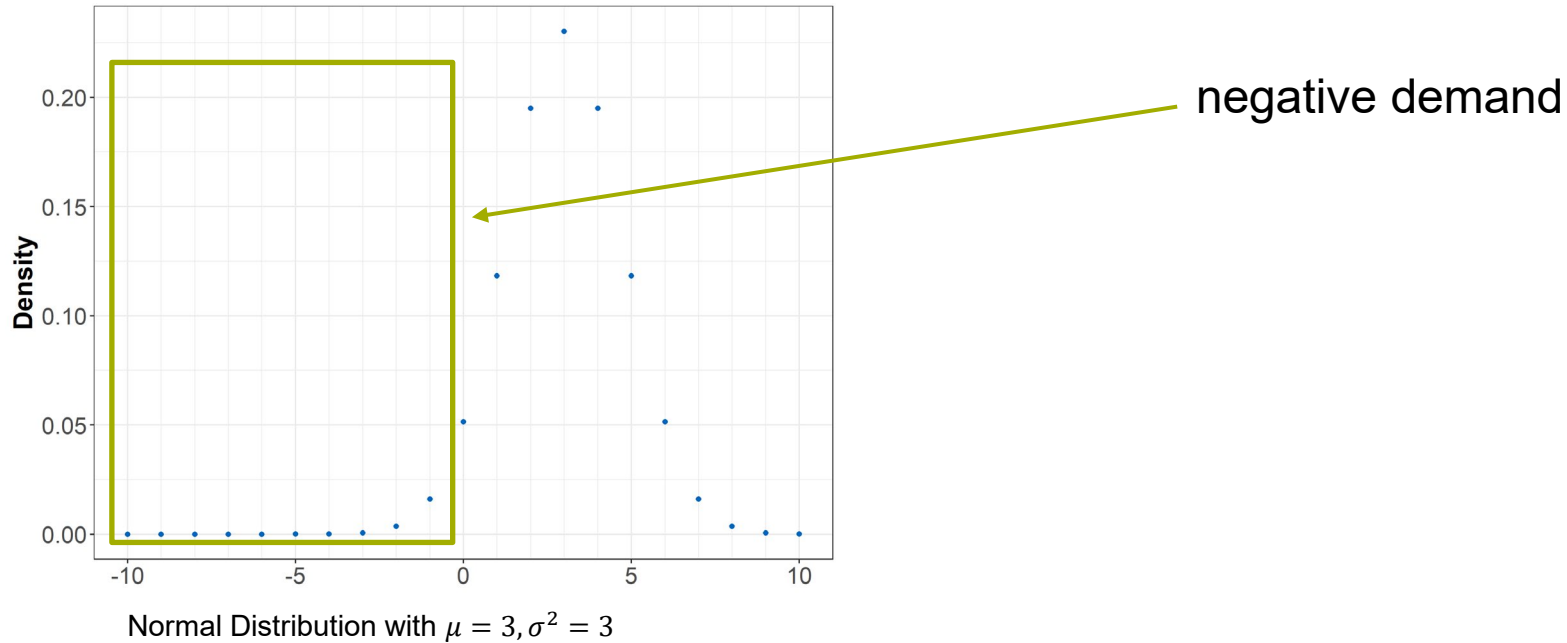


Normal Distribution with $\mu = 250, \sigma^2 = 250$

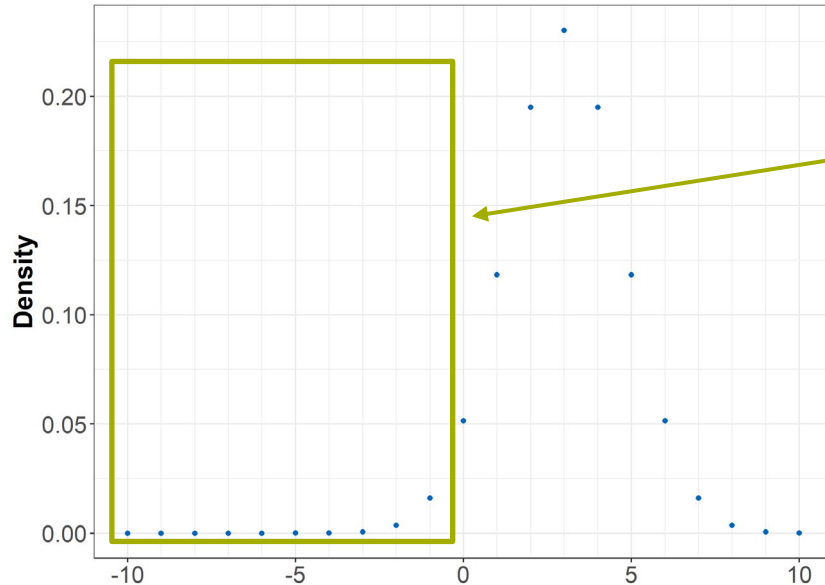
Is demand normally distributed?



Is demand normally distributed?



Is demand normally distributed?



Normal Distribution with $\mu = 3, \sigma^2 = 3$

negative demand

If X is normally distributed with mean μ and standard deviation σ , then $\frac{X-\mu}{\sigma}$ is standard normally distributed and

$$P(X \leq 0) = F_{0,1}\left(\frac{0-\mu}{\sigma}\right) = F_{0,1}\left(-\frac{\mu}{\sigma}\right).$$

$$F_{0,1}(-3) = 0.0013$$

$$F_{0,1}(-4) = 0.000032$$

To use the normal distribution, it good to have $4\mu > \sigma$.

Approximate demand with a normal random variable?

Demand can be approximated with a normal random variable if,

- units are uncountable (e.g., demand for water or oil),
- numbers are relatively high, so that rounding is less of an issue (e.g., $\mu > 100$),
- $\frac{\mu}{\sigma}$ ratio is high enough (e.g., rule of thumb, above 4),
- if $\sigma^2 \approx \mu$ (that is, Poisson distribution is likely unsuitable).

Approximate demand with a uniform random variable?

Demand can be approximated with a uniform random variable if,

- outliers are highly likely,
- rather a one-shot game: e.g., product launch and “anything can happen” scenario,
- we only need a very rough approximation,
- demand is just one source of uncertainty besides, e.g., yield and supply uncertainty, and we are interested in closed-form, explicit solutions.

The Art of Modeling Real Life

